Introduction to SIEM

Log Centralization, Normalization, and Threat Detection

Security Information and Event Management (SIEM) systems serve as the centralized "brain" of a Security Operations Center (SOC), collecting and analyzing logs from diverse network sources. This module details the distinction between **Host-Centric** and **Network-Centric** log sources and addresses the inherent challenges of fragmented data analysis. Key SIEM functionalities—such as **Normalization**, **Parsing**, and **Correlation** are explained as the primary methods for transforming raw data into actionable intelligence.

We explore log ingestion techniques including **Agents** and **Syslog**, alongside the logic behind real-time detection rules based on specific **Event IDs**.

By mastering these concepts, analysts can effectively investigate alerts, distinguish between true and false positives, and maintain unified visibility across the entire organizational infrastructure.

This **documentation** and **learning** approach is strongly inspired by the structured, hands-on content provided by **TryHackMe**, which significantly contributed to practical understanding of "SIEM" Fundamentals.

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1. Log Sources and Classification

In a modern network, every device acts as a **Log Source**, creating a digital trail of activity. These are broadly categorized into two types:

A. Host-Centric Log Sources

These capture events occurring within a specific endpoint (e.g., Windows/Linux workstations, servers).

- **Examples:** File access, user authentication attempts, process executions, **PowerShell** scripts, and registry modifications.

B. Network-Centric Log Sources

These are generated when hosts communicate across the network or the Internet.

- **Examples: SSH** connections, **VPN** access, web traffic, and firewall/IDS/IPS logs.

2. The Necessity of SIEM

Without a SIEM, analysts face significant "Answers Nowhere" challenges:

- **Fragmentation:** Hundreds of events per second are scattered across different devices.
- **No Centralization:** Manually connecting to each device (via **SSH** or **RDP**) is inefficient.
- **Limited Context:** Individual logs look harmless in isolation but reveal attacks when correlated.
- **Format Issues:** Every manufacturer uses different log formats, making manual reading impossible.

3. Core Features of a SIEM Solution

A SIEM transforms raw data into a manageable and actionable format through the following processes:

Feature	Description	Beginner Example
Centralization	Pulls all logs into one single database.	Viewing Windows and Linux logs on the same screen.
Normalization	Converts different log formats into a consistent standard.	Mapping "LogonID" (Windows) and "uid" (Linux) to a single "User" field.
Parsing	Breaking down raw log strings into searchable fields.	Extracting the **IP Address** from a long web server log line.
Correlation	Linking related events across different sources.	Connecting a failed **VPN** login to a subsequent **PowerShell** execution.
Alerting	Triggering notifications when specific logic is met.	Alerting if a user fails a login 5 times in 10 seconds.

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4. Log Ingestion Methods

SIEM solutions use various techniques to bring data into the system:

- **Agent / Forwarder:** A lightweight tool (like a **Splunk Forwarder**) installed on the endpoint to push logs to the server.
- **Syslog:** A standard protocol used by routers and web servers to send real-time data.
- **Manual Upload:** Ingesting offline files (CSV, JSON) for historical analysis.
- **Port-Forwarding:** Configuring endpoints to send data directly to a specific port on the SIEM instance.

5. Alerting Process and Analysis

SIEM systems use **Detection Rules** to identify threats. These rules are logical expressions that monitor field-value pairs.

Key Event IDs for Detection

Event ID	Activity	Logic for Alert
104	**Event Log Cleared**	If LogSource = Windows AND EventID = 104 $\rightarrow$ **Alert**.
4688	**Process Execution**	If EventID = 4688 AND Process = whoami $\rightarrow$ **Alert**.

Analysis Outcomes

- **True Positive:** A real security threat is identified; requires immediate response/isolation.
- **False Positive:** A benign activity triggered the rule; requires **Rule Tuning** to reduce noise.

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6. Practical Lab Work

I researched the SIEM dashboard provided in the simulation and **found** a series of events matching pre-configured security rules. Below is how **I solved** the investigation:

- **Work Done: I practiced** monitoring the real-time event flow. After clicking the **Start Suspicious Activity** button, **I found** an alert was triggered by the execution of a specific process.
- **Solution: I researched** the event logs and **found** that the process cmd.exe was used to execute the whoami command. **I practiced** identifying the user responsible, which **I founded** to be **"Chris"**.
- **Host Identification: I founded** that the activity occurred on the hostname **"DESKTOP-5471"**.
- **Analysis:** Since the process matched the "Unauthorized Command" rule, **I solved** the triage by classifying this as a **True Positive**.
- **Result: I founded** the flag THM{000_INTRO_TO_SOC} after successfully executing the remediation action.

Conclusion and Cheatsheet

Conclusion

SIEM is the foundation of modern security operations, bridging the gap between raw data and actionable intelligence. By centralizing logs, normalizing formats, and correlating disparate events, SIEM allows SOC teams to detect complex attack patterns that would otherwise remain hidden. Proficiency in managing **Log Ingestion** and refining **Detection Rules** is essential for minimizing false positives and ensuring a rapid response to legitimate security incidents.

Fast Revision Cheatsheet

- **Host-Centric:** Focuses on internal OS activity (Registry, Files).
- **Network-Centric:** Focuses on communication (Firewall, VPN).
- **Parsing:** Extracting specific data pieces from a raw log.
- **Normalization:** Making all logs follow the same format.
- **Event ID 104:** Critical alert for "Logs Cleared" (Post-Exploitation).
- **Event ID 4688:** Standard ID for Process Creation/Execution.
- **Splunk/ELK:** Popular SIEM industry examples.
- **Rule Tuning:** The process of adjusting a rule to stop **False Positives**.
- **Forwarder:** The agent used to send logs from an endpoint to the SIEM.